

Robust Decision Making for Autonomous Vehicles at Highway On-Ramps: A Constrained Adversarial Reinforcement Learning Approach

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Abstract—Reinforcement learning has demonstrated its potential in a series of challenging domains. However, many real-world decision making tasks involve unpredictable environmental changes or unavoidable perception errors that are often enough to mislead an agent into making suboptimal decisions and even cause catastrophic failures. In light of these potential risks, reinforcement learning with application in safety-critical autonomous driving domain remains tricky without ensuring robustness against environmental uncertainties (e.g., road adhesion changes or measurement noises). Therefore, this paper proposes a novel constrained adversarial reinforcement learning approach for robust decision making of autonomous vehicles at highway on-ramps. Environmental disturbance is modelled as an adversarial agent that can learn an optimal adversarial policy to thwart the autonomous driving agent. Meanwhile, observation perturbation is approximated to maximize the variation of the perturbed policy through a white-box adversarial attack technique. Furthermore, a constrained adversarial actor-critic algorithm is presented to optimize an on-ramp merging policy while keeping the variations of the attacked driving policy and action-value function within bounds. Finally, the proposed robust highway on-ramp merging decision making method of autonomous vehicles is evaluated in three stochastic mixed traffic flows with different densities, and its effectiveness is demonstrated in comparison with the competitive baselines.

Index Terms—Autonomous vehicle, ramp merging, robust decision making, reinforcement learning, adversarial attack.

I. INTRODUCTION

AUTONOMOUS vehicles have attracted considerable attention from academia and industry [1], [2]. Their potential societal benefits stem from safer transportation with fewer crashes to reduced congestion and lower emissions [3]. However, the safety of autonomous driving remains major concerns of researchers, venture capitalists and the general public [4], [5]. Undesirable decisions by autonomous vehicles may cause a heavy human and economic toll. It is obvious that there is still a long way to go to meet the strict requirements and high expectations associated with autonomous vehicles. Reinforcement learning (RL) has achieved impressive success

in a range of complex decision making tasks (e.g., Go, StarCraft) [6], [7], [8]. Therefore, RL with application in decision making system of autonomous vehicles has become one of hot research topics [9], [10].

RL based decision making of autonomous driving can be broadly divided into three categories: lateral, longitudinal and coordinated decision making. RL based lateral decision making methods of autonomous vehicles generally leverage RL algorithm to learn lane change policies or determine target lanes [11], [12], [13].

RL based longitudinal decision making approaches usually employ RL algorithm to determine the autonomous driving speed modes (e.g., keeping, acceleration and deceleration). A longitudinal decision making framework of autonomous vehicles at intersection scenarios is proposed via safe RL relying on a model-checker and a learned belief updater in [14]. A longitudinal decision making system with double deep Q-network (DQN) is designed to learn speed mode which includes keeping, acceleration and deceleration in [15] and [16]. An emergency braking decision making approach is developed for autonomous vehicles via deep deterministic policy gradient (DDPG) in [17]. A DDPG based longitudinal decision making for autonomous driving is presented to train car-following model in [18]. A hierarchical RL framework for autonomous driving longitudinal decision making is proposed to determine sub-goal and generate acceleration or deceleration choices in [19]. The longitudinal motion modes of autonomous vehicles can be determined through combining dueling DQN with transfer learning in [20].

RL based coordinated decision making methods mainly adopted RL algorithm to determine autonomous driving longitudinal and lateral behaviors simultaneously. A longitudinal and lateral coordinated decision making method with AlphaGo Zero algorithm is advanced for autonomous driving in [21]. The requested speed and target lane can be determined using the five decision making behaviors simultaneously. A coordinated decision making scheme with DQN is developed, which can learn discrete speed modes and lane change behaviors of autonomous vehicles in [22]. A coordinated decision making method based on DDPG algorithm is presented to learn autonomous driving throttle and steering maneuvers in [23]. An optimization embedded RL based on actor-critic framework is proposed to learn longitudinal and lateral coordinated driving policies in [24].

On-ramp merging is a challenging task. Inappropriate merging maneuvers may cause typical consequences, e.g.,

Manuscript received 26 February 2022; revised 11 July 2022 and 29 September 2022; accepted 12 December 2022. Date of publication 20 December 2022; date of current version 29 March 2023. This work was supported in part by the A*STAR AME Young Individual Research Grant A2084c0156; in part by the SUG-NAP Grant of Nanyang Technological University, Singapore; and in part by the National Research Foundation (NRF) under Project NRF2021-NRF-ANR003 HM Science. The Associate Editor for this article was J. E. Naranjo. (*Corresponding author: Chen Lv.*)

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Digital Object Identifier 10.1109/TITS.2022.3229518

congestion, collision, or time-consuming. Numerous studies attempt to solve the on-ramp merging problem of autonomous vehicles via RL in recent years. An autonomous driving on-ramp merging decision making approach by controlling acceleration and steering is developed using a DQN based on long short-term memory (LSTM) architecture in [25]. A DQN with partially observable Markov decision process (POMDP) is introduced to train an autonomous driving RL agent for on-ramp merging in dense traffic in [26]. A DDPG based on multi-objective reward function is presented to learn a highway on-ramp merging policy for autonomous vehicles in [27]. An interpretable longitudinal decision making for autonomous vehicles is advanced to determine throttle or brake mode via a latent space actor-critic algorithm in [28]. An autonomous driving on-ramp merging scheme is developed by combining DDPG with model predictive control (MPC) in [29].

However, many real-world autonomous driving problems involve unpredictable environmental changes or unavoidable perception errors that may mislead an autonomous vehicle into taking unsafe actions and even cause serious traffic accidents. In view of these potential dangers, autonomous vehicles are required to ensure that their decision making systems are robust against environmental uncertainties. Notwithstanding, the RL based robust decision making has not been fully explored in autonomous driving domain. Additionally, few researches simultaneously consider and cope with environmental disturbances (e.g., road adhesion changes or mass variation) and observation perturbations (e.g., measurement noises or perception errors). While a few existing studies have attempted to solve ramp metering tasks considering uncertainties via macroscopic robust traffic flows model [30], [31], our work has a couple of differences or advantages compared to these methods. Firstly, the above researches aim to minimize the travel delay of macroscopic traffic flows with uncertainties in demand or supply. In contrast, this work focuses on the decision making technique of autonomous vehicles, which tries to enable autonomous driving agents to learn robust, safe and efficient on-ramp merging policies against adversarial environmental uncertainties. Secondly, these approaches to tackle the macroscopic ramp metering problem do not consider the interactions between agents, which makes them difficult to deploy in real-world applications. For instance, collisions between vehicles on a ramp and vehicles on a highway may occur, but it is intractable for these methods to address this problem. By comparison, our research enables autonomous vehicles at highway on-ramps to learn safe decision making behaviors in online interactions with traffic participants. Thirdly, the generalization of these schemes in unseen scenarios remains challenging, since they require solving an optimization problem in each different situation. By contrast, this study allows trained agents to deal with scenarios that have not been seen before without learning or optimizing. For example, our autonomous driving agents are trained in traffic flows with a specified density, and the trained agents can then be deployed directly to other traffic flows with different densities for testing.

Here, this paper presents a novel constrained adversarial reinforcement learning (CARL) approach for robust decision

making of autonomous vehicles at highway on-ramps. The main contributions of this paper are summarized as follows:

- The proposed CARL method enables agents to learn robust on-ramp merging policies against adversarial environmental uncertainties and ensure driving safety and travel efficiency.
- Two kinds of environment uncertainties are simultaneously considered to simulate the worst-case situations. The environmental disturbance is modelled as an adversarial agent through two-player zero-sum Markov games. The adversary (i.e., environmental disturbance) can learn an optimal adversarial policy to thwart the protagonist (i.e., autonomous driving agent). Meanwhile, the observation perturbation is approximated to maximize the variation of the perturbed policy via a white-box adversarial attack technique.
- A constrained adversarial actor-critic (CAAC) algorithm is advanced to optimize the on-ramp merging policy while minimizing the variations of the driving policy and action-value function attacked by environmental disturbances and observation perturbations.

Three testing cases based on different traffic flow densities are carried out to evaluate the performance of the proposed robust highway on-ramp merging decision making scheme by simulation of urban mobility (SUMO) [32], [33]. The results demonstrate that our approach is effective and outperforms the competitive baselines.

The remainder is organized as follows. Preliminaries are introduced in Section II. Section III illustrates the framework of the proposed method. Our CARL algorithm is elaborated in Section IV. The experimental results are discussed in Section V. The conclusion is drawn in Section VI.

II. PRELIMINARIES

A. Markov Decision Process

A Markov Decision Process (MDP) provides a standard mathematical formalism for modeling decision making problems. An MDP can be represented by the 5-tuple $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$ with state space $\mathcal{S}$, action space $\mathcal{A}$, state transition probability $p : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$, reward function $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ and discount factor $\gamma \in (0, 1)$.

The goal of an MDP is to learn a optimal policy π^* which can maximize the cumulative discounted reward. This setting can be written as:

$$\pi^* \in \arg \max_{\pi} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t) \right], \quad (1)$$

where π represents policy, t and T are time steps, s_t and a_t denotes state and action at time step t .

B. Two-Player Zero-Sum Markov Games

In this paper, two-player zero-sum Markov games are leveraged to model the environmental disturbance as the adversary that thwarts the protagonist. This games describe a situation which involves two players, where the payoff of a player equals the loss of the opponent [34]. In this games, protagonist

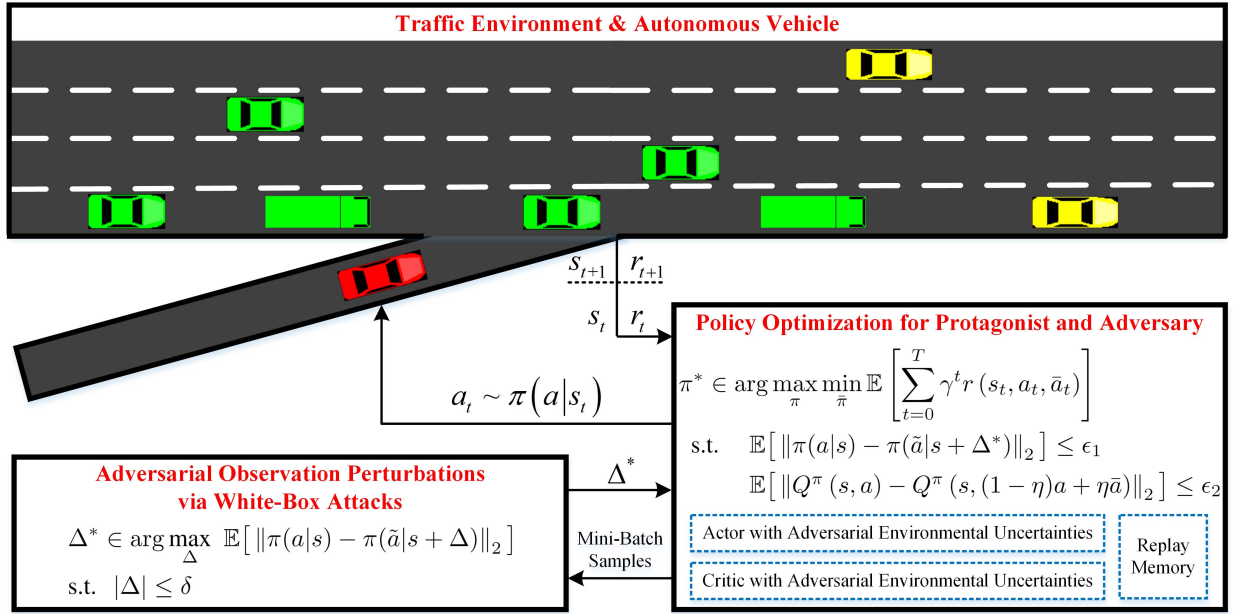


Fig. 1. Illustration of the proposed robust highway on-ramp merging decision making framework for autonomous vehicles.

and adversary take actions a and $\bar{a}$ based on policies π and $\bar{\pi}$ respectively, and state transition probability $p(s'|s, a, \bar{a})$ depend on both players. s and s' represent state and next state respectively.

The value function with π and $\bar{\pi}$ can be expressed as $v^{\pi, \bar{\pi}}(s) \equiv \mathbb{E}[\sum_{t=0}^T \gamma^t r(s_t, a_t, \bar{a}_t)]$. The protagonist tries to maximize the value function (i.e., total expected discounted reward), and the adversary attempts to minimize this function. Nash equilibrium is the most common approach to define the solution of a non-cooperative game involving two or more players. A Nash equilibrium $(\pi^*, \bar{\pi}^*)$ in two-person zero-sum Markov games exists when the following relation holds [35]:

$$\begin{aligned} v^{\pi^*, \bar{\pi}^*}(s) &= \max_{\pi} \min_{\bar{\pi}} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t, \bar{a}_t) \right] \\ &= \min_{\bar{\pi}} \max_{\pi} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t, \bar{a}_t) \right], \end{aligned} \quad (2)$$

where π^* and $\bar{\pi}^*$ represent the optimal policies of protagonist and adversary respectively, v^* is optimal equilibrium value in the games. The existence of v^* , π^* and $\bar{\pi}^*$ is discussed and proved for this games in [36] and [37].

One way to solve two-person zero-sum Markov is via discretizing the continuous state and action spaces and using dynamic programming. However, the complexity of the greedy solution to find the Nash equilibria is exponential in the cardinality of the action spaces, which makes it intractable [38]. In addition, the majority of Markov game methods require solving the equilibrium solution of a multiplayer value at each iteration, which is a typically tricky minimax optimization problem [39]. In contrast, we focus on approximating the equilibrium solution by alternately optimizing the adversary and protagonist.

III. ROBUST HIGHWAY ON-RAMP MERGING DECISION MAKING FRAMEWORK FOR AUTONOMOUS VEHICLES

A block diagram of the proposed robust highway on-ramp merging decision making framework for autonomous driving is shown in Fig. 1.

The adversary agent and the white-box adversarial attack technique are employed to simulate or generate the adversarial environmental disturbances and observation perturbations respectively. The RL based autonomous driving agent (protagonist) seeks to maximize the expected return under adversarial environmental uncertainties while satisfying the constraint setting. In Fig. 1, Δ represents observation perturbation, and Δ^* is the optimal adversarial observation perturbation. δ denotes the desired bounds of the observation perturbation. The policy $\pi(a|s)$ describes which action a to take in state s . $\bar{a}$ represents the action taken by the protagonist under observation perturbations. ϵ_1 and ϵ_2 are expected minimum deviations respectively. η denotes the adversarial weight considering the adversary. $Q^\pi(\cdot)$ represents on-policy action-value function.

Additionally, the output of the autonomous driving agent is continuous longitudinal acceleration. In other words, the longitudinal acceleration or deceleration of an autonomous vehicle is determined to implement highway on-ramp merging decision making via the proposed CARL approach.

IV. CONSTRAINED ADVERSARIAL REINFORCEMENT LEARNING

A. Constrained Adversarial Markov Decision Process

In this section, a constrained adversarial Markov decision process (CAMDP) is proposed to model the decision making behaviors of RL agent under environmental uncertainty and constraint setting. A CAMDP can be regarded as special two-person zero-sum Markov games between the protagonist and the adversary.

Definition 1: A CAMDP is characterized by a 7-tuple $(\mathcal{S}, \mathcal{A}^{\text{mix}}, p, r, c, \Delta, \gamma)$. $\mathcal{A}^{\text{mix}}$ is a mixed action space which contains the actions taken by the protagonist and the adversary. $p : \mathcal{S} \times \mathcal{A}^{\text{mix}} \times \mathcal{S} \rightarrow \mathbb{R}$ is the state transition probability which is a mapping from the current state s and the mixed action a^{mix} to the probability distribution over the state space $\mathcal{S}$. $r : \mathcal{S} \times \mathcal{A}^{\text{mix}} \rightarrow \mathbb{R}$ denotes reward function under the mixed action space, and $c : \mathcal{S} \times \mathcal{A}^{\text{mix}} \rightarrow \mathbb{R}$ represents constraint function with the mixed action a^{mix} . Δ represents observation perturbation.

The CAMDP attempts to solve the following problem:

$$\begin{aligned} \max_{\pi} \min_{\bar{\pi}} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t^{\text{mix}}) \right], \\ \text{s.t. } \mathbb{E} [c(s, a^{\text{mix}}, \Delta)] \leq \epsilon, \end{aligned} \quad (3)$$

where ϵ is an predefined minimum deviation, and the mixed action a^{mix} is defined as:

$$a^{\text{mix}} \equiv (1 - \eta)a + \eta\bar{a}. \quad (4)$$

B. White-Box Adversarial Attacks

The white-box adversarial attack technique is implemented to approximate the optimal adversarial observation perturbations in this section. This attack technique requires the complete knowledge of the objective function to find successful adversarial examples [40], and it can be formulated as:

$$\begin{aligned} \Delta^* \in \arg \max_{\Delta} \mathbb{E} [\|\pi(a|s) - \pi(\bar{a}|s + \Delta)\|_2], \\ \text{s.t. } |\Delta| \leq \delta, \end{aligned} \quad (5)$$

where δ is the bound of observation perturbation.

To simplify the above constrained optimization problem, a hyperbolic tangent function $\tanh(\cdot)$ is adopted to limit the range of adversarial observation perturbation:

$$\Delta(x) = k \frac{e^x - e^{-x}}{e^x + e^{-x}}, \quad (6)$$

where x is the optimization variable, k is an upper bound, and e is the natural logarithm. Moreover, the constrained optimization problem (5) can be rewritten as:

$$x^* \in \arg \max_x \mathbb{E} [\|\pi(a|s) - \pi(\bar{a}|s + \Delta(x))\|_2], \quad (7)$$

where x^* is the optimal solution of this optimization problem.

Hence, the optimal adversarial observation perturbation Δ^* can be obtained:

$$\Delta^* = k \frac{e^{x^*} - e^{-x^*}}{e^{x^*} + e^{-x^*}}. \quad (8)$$

Algorithm (1) outlines the proposed white-box adversarial attack scheme.

Algorithm 1 White-Box Adversarial Attack Technique

Input: Initial optimization variable x_0 , the learning rate α .

for $i = 1, 2, \dots, I$ **do**

 Update the optimization variable:

$$x_{i+1} \leftarrow x_i + \alpha \nabla_{x_i} \mathbb{E} [\|\pi(a|s) - \pi(\bar{a}|s + \Delta(x_i))\|_2].$$

end for

Output: $\Delta(x_{I+1})$.

C. Constrained Adversarial Actor-Critic

In this section, the proposed CAAC algorithm is introduced to learn the robust optimal highway on-ramp merging policies.

CAAC seeks to solve the following constrained optimization problem:

$$\begin{aligned} \pi^* \in \arg \max_{\pi} \min_{\bar{\pi}} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t^{\text{mix}}) \right], \\ \text{s.t. } \begin{aligned} c_1(s, \Delta^*) &\leq \epsilon_1, \\ c_2(s, a^{\text{mix}}) &\leq \epsilon_2, \end{aligned} \end{aligned} \quad (9)$$

where the constraint functions $c_1(\cdot)$ and $c_2(\cdot)$ can be defined as:

$$c_1(s, \Delta^*) \equiv \mathbb{E} [\|\pi(a|s) - \pi(\bar{a}|s + \Delta^*)\|_2], \quad (10)$$

$$c_2(s, a^{\text{mix}}) \equiv \mathbb{E} [\|Q^\pi(s, a) - Q^\pi(s, a^{\text{mix}})\|_2]. \quad (11)$$

A policy iteration (PI) algorithm is developed to approximate CAMDP, which is named after constrained adversarial PI (CAPI). CAPI method consists of policy evaluation and policy improvement, and these two learning processes are updated alternately until the policy converges.

To solve the complicated nonlinear constrained optimization task (9), we simplify and decompose the original problem. Therefore, the following two optimization tasks can be constructed:

$$\min_{\bar{\pi}} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t^{\text{mix}}) \right], \quad (12)$$

$$\begin{aligned} \max_{\pi} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t^{\text{mix}}) \right], \\ \text{s.t. } \begin{aligned} c_1(s, \Delta^*) &\leq \epsilon_1, \\ c_2(s, a^{\text{mix}}) &\leq \epsilon_2. \end{aligned} \end{aligned} \quad (13)$$

The Lagrangian of the constrained optimization problem (13) can be written as:

$$\begin{aligned} L(\pi, \bar{\pi}, \lambda_1, \lambda_2) = \mathbb{E} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t^{\text{mix}}) + \lambda_1 (\epsilon_1 - c_1(s, \Delta^*)) \right. \\ \left. + \lambda_2 (\epsilon_2 - c_2(s, a^{\text{mix}})) \right], \end{aligned} \quad (14)$$

where λ_1 and λ_2 are dual variables, $\lambda_1 \geq 0$, $\lambda_2 \geq 0$.

1) *Constrained Adversarial Policy Evaluation*: In this paper, an on-policy action-value function considering the attacks of the adversary is defined as:

$$Q^{\pi, \bar{\pi}}(s, a^{\text{mix}}) \equiv (1 - \eta)Q^{\pi}(s, a) + \eta Q^{\pi}(s, a^{\text{mix}}). \quad (15)$$

The action-value function $Q^{\pi, \bar{\pi}}(s_t, a_t^{\text{mix}})$ at time step t can be computed under fixed policies of the protagonist and the adversary iteratively by a modified Bellman backup operator $\mathcal{T}^{\text{mix}}$:

$$\mathcal{T}^{\text{mix}} Q^{\pi, \bar{\pi}}(s_t, a_t^{\text{mix}}) \equiv r(s_t, a_t) + \gamma \mathbb{E}[Q^{\pi, \bar{\pi}}(s_{t+1}, a_{t+1}^{\text{mix}})]. \quad (16)$$

CAAC algorithm employs two parameterized action-value functions with network parameters ϕ^z , $z \in \{1, 2\}$ to speed up model training. The parameters of action-value function can be learned through minimizing the following loss function of critic network:

$$J_c(\phi^z) = \mathbb{E}_{T_s \sim \mathcal{D}} \left[\|y - Q^{\pi}(s_t, a_t; \phi^z)\|_2^2 \right], \quad (17)$$

where T_s represents transition sampled from replay buffer $\mathcal{D}$, and y is the target value of the action-value function at the time step t .

To avoid overfitting policy model to narrow peaks in the action-value function, a smoothing regularization method [41] is utilized via adding a small amount of random noises to the selected action based on the deterministic policy. Hence, with the parameterized deterministic policy π , the action with the random noises at time step t can be represented as:

$$\hat{\pi}(s_t; \theta) = \pi(s_t; \theta) + \tilde{k}\beta, \quad (18)$$

$$\beta \sim \mathcal{N}(0, 1), \quad (19)$$

where θ is the parameter of actor network, $\tilde{k}$ is a weight coefficient, and $\mathcal{N}(\cdot)$ denotes Gaussian distribution.

Furthermore, the mixed action at time step t can be redefined:

$$a_t^{\text{mix}} \equiv (1 - \eta)\hat{\pi}(s_t; \theta) + \eta\bar{\pi}(s_t; \bar{\theta}), \quad (20)$$

where $\bar{\theta}$ represents the network parameter of the adversary.

With Eq. (15), target parameterized action-value function with considering the attacks of the adversary at time step t is defined as:

$$\begin{aligned} \hat{Q}^{\pi, \bar{\pi}}(s_t, a_t^{\text{mix}}) &\equiv (1 - \eta)\hat{Q}^{\pi}(s_t, \hat{\pi}(s_t; \theta); \bar{\phi}^z) \\ &+ \eta\hat{Q}^{\pi}(s_t, a_t^{\text{mix}}; \bar{\phi}^z), \end{aligned} \quad (21)$$

where $\bar{\phi}^z$ is the network parameter of the target action-value function, and $z \in \{1, 2\}$.

To fend off overestimation in the action-value function, the minimum estimation among two target parameterized action-value functions is leveraged to train critic network. With

Eq. (16), Eq. (20) and Eq. (21), y can be written as:

$$\begin{aligned} y &= r(s_t, a_t) + \gamma \min_{z \in \{1, 2\}} \hat{Q}^{\pi, \bar{\pi}}(s_{t+1}, a_{t+1}^{\text{mix}}) \\ &= r(s_t, a_t) + \gamma \min_{z \in \{1, 2\}} \left[(1 - \eta)\hat{Q}^{\pi}(s_{t+1}, \hat{\pi}(s_{t+1}; \theta); \bar{\phi}^z) \right. \\ &\quad \left. + \eta\hat{Q}^{\pi}(s_{t+1}, a_{t+1}^{\text{mix}}; \bar{\phi}^z) \right] \\ &= r(s_t, a_t) + \gamma \min_{z \in \{1, 2\}} \left[(1 - \eta)\hat{Q}^{\pi}(s_{t+1}, \hat{\pi}(s_{t+1}; \theta); \bar{\phi}^z) \right. \\ &\quad \left. + \eta\hat{Q}^{\pi}(s_{t+1}, (1 - \eta)\hat{\pi}(s_{t+1}; \theta) + \eta\bar{\pi}(s_{t+1}; \bar{\theta}); \bar{\phi}^z) \right]. \end{aligned} \quad (22)$$

The network parameters $\bar{\phi}^z$ of the target action-value functions can be updated using polyak averaging:

$$\bar{\phi}^z \leftarrow \rho \bar{\phi}^z + (1 - \rho)\phi^z, \quad (23)$$

where ρ is a scale coefficient between 0 and 1.

2) *Constrained Adversarial Policy Improvement*: In CAPI, the goal of policy improvement is to update and optimize the policies of RL agent. The RL agent seeks to maximize the expected return of the policy while satisfying the constraint setting.

According to Lagrange duality theory [42] and Eq. (14), the Lagrange dual function of the constrained optimization problem (13) can be derived as:

$$\begin{aligned} \bar{L}(\lambda_1, \lambda_2) &= \max_{\pi} L(\pi, \bar{\pi}, \lambda_1, \lambda_2) \\ &= \max_{\pi} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t^{\text{mix}}) + \lambda_1(\epsilon_1 - c_1(s, \Delta^*)) \right. \\ &\quad \left. + \lambda_2(\epsilon_2 - c_2(s, a^{\text{mix}})) \right]. \end{aligned} \quad (24)$$

Moreover, the Lagrange dual problem associated with the problem (13) can be expressed as:

$$\begin{aligned} \min_{\lambda_1, \lambda_2} \bar{L}(\lambda_1, \lambda_2) &= \min_{\lambda_1, \lambda_2} \max_{\pi} L(\pi, \bar{\pi}, \lambda_1, \lambda_2) \\ &= \min_{\lambda_1, \lambda_2} \max_{\pi} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t^{\text{mix}}) \right. \\ &\quad \left. + \lambda_1(\epsilon_1 - c_1(s, \Delta^*)) + \lambda_2(\epsilon_2 - c_2(s, a^{\text{mix}})) \right]. \end{aligned} \quad (25)$$

Therefore, according to Eq. (12) and Eq. (25), the optimal adversarial policy $\bar{\pi}^*$, the optimal learning policy π^* , the optimal dual variables λ_1^* and λ_2^* can be approximated iteratively. First given fixed π , optimize adversarial policy $\bar{\pi}$ by minimizing the expected return. Then provide the optimal adversarial policy $\bar{\pi}^*$, fixed λ_1 and λ_2 , then optimize the learning policy π via maximizing the Lagrangian respectively. Additionally, plug in $\bar{\pi}^*$ and π^* , then find λ_1^* and λ_2^* by minimizing $L(\bar{\pi}^*, \pi^*, \lambda_1, \lambda_2)$. Hence, with Eq. (25), the following expressions can be derived:

$$\bar{\pi}^* = \arg \min_{\bar{\pi}} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t^{\text{mix}}) \right], \quad (26)$$

$$\pi^* = \arg \max_{\pi} L(\bar{\pi}^*, \pi, \lambda_1, \lambda_2), \quad (27)$$

$$\lambda_1^*, \lambda_2^* = \arg \min_{\lambda_1, \lambda_2} L(\bar{\pi}^*, \pi^*, \lambda_1, \lambda_2). \quad (28)$$

To reduce the estimation error of the expected return based on π and $\bar{\pi}$, the double $Q^{\pi, \bar{\pi}}(\cdot)$ trick is leveraged. With Eq. (15) and Eq. (20), the average action-value function with π and $\bar{\pi}$ at time step t can be represented as:

$$\begin{aligned} \tilde{Q}^{\pi, \bar{\pi}}(s_t, a_t^{\text{mix}}) &= \frac{1}{2} [Q^{\pi, \bar{\pi}}(s_t, a_t^{\text{mix}}; \phi^1) + Q^{\pi, \bar{\pi}}(s_t, a_t^{\text{mix}}; \phi^2)] \\ &= \frac{1}{2} [(1 - \eta)(Q^\pi(s_t, \hat{\pi}(s_t; \theta); \phi^1) + Q^\pi(s_t, \hat{\pi}(s_t; \theta); \phi^2)) \\ &\quad + \eta(Q^\pi(s_t, a_t^{\text{mix}}; \phi^1) + Q^\pi(s_t, a_t^{\text{mix}}; \phi^2))] \\ &= \frac{1}{2} [(1 - \eta)(Q^\pi(s_t, \hat{\pi}(s_t; \theta); \phi^1) + Q^\pi(s_t, \hat{\pi}(s_t; \theta); \phi^2)) \\ &\quad + \eta(Q^\pi(s_t, (1 - \eta)\hat{\pi}(s_t; \theta) + \eta\bar{\pi}(s_t; \bar{\theta}); \phi^1) \\ &\quad + Q^\pi(s_t, (1 - \eta)\hat{\pi}(s_t; \theta) + \eta\bar{\pi}(s_t; \bar{\theta}); \phi^2))]. \end{aligned} \quad (29)$$

The adversary attempts to minimize the expected return, and the optimal adversarial policy can be approximated via minimizing the following loss function of adversary network:

$$J_{\bar{\pi}}(\bar{\theta}) = \mathbb{E}_{T s \sim \mathcal{D}} [\tilde{Q}^{\pi, \bar{\pi}}(s_t, a_t^{\text{mix}})]. \quad (30)$$

The protagonist seeks to maximize the expected return while minimizing the nonlinear constraints, and the optimal learning policy can be solved via maximizing the following objective function of actor network:

$$J_{\pi}(\theta) = \mathbb{E}_{T s \sim \mathcal{D}} [\tilde{Q}^{\pi, \bar{\pi}}(s_t, a_t^{\text{mix}}) - \lambda_1 c_1(s_t, \Delta^*) - \lambda_2 c_2(s_t, a_t^{\text{mix}})]. \quad (31)$$

Additionally, with Eq. (28), the dual variables can be updated via minimizing the following loss function:

$$J_d(\lambda_1, \lambda_2) = \mathbb{E}_{T s \sim \mathcal{D}} [\lambda_1(\epsilon_1 - c_1(s_t, \Delta^*)) + \lambda_2(\epsilon_2 - c_2(s_t, a_t^{\text{mix}}))]. \quad (32)$$

D. Algorithm Implementation

Algorithm 2 outlines the proposed CARL algorithm in detail. d_t indicates if s_{t+1} is terminal. ω represents a delayed update coefficient. The proposed scheme can optimize autonomous driving RL agent through the following rough procedure. The initial the network parameters of adversary, actor and critic are sampled from a random distribution. The initial target action-value function network parameters can be obtained by $\bar{\phi}^1 \leftarrow \phi^1$ and $\bar{\phi}^2 \leftarrow \phi^2$. In addition, dual variables λ_1 and λ_2 are initialized randomly. For each iteration, RL agent first need to collect the data of T timesteps and store them in buffer $\mathcal{D}$. The next state and the reward function can be generated via the environment. The optimal adversarial observation perturbations Δ^* are approximated via the white-box adversarial attack technique. Then the network parameters of adversary, actor and critic can be learned with Eq. (30), Eq. (31) and Eq. (17) respectively. Furthermore, the target action-value function network parameters, dual variables λ_1 and λ_2 can be updated with Eq. (23) and Eq. (32) respectively.

Algorithm 2 Constrained Adversarial Reinforcement Learning

- 1: Initialize adversary network parameters $\bar{\theta}$, actor network parameters θ , critic network parameters ϕ^1 and ϕ^2 .
- 2: Initialize target action-value function network parameters $\bar{\phi}^1 \leftarrow \phi^1$, and $\bar{\phi}^2 \leftarrow \phi^2$.
- 3: Initialize dual variables λ_1 and λ_2 .
- 4: Initialize an empty replay buffer $\mathcal{D}$.
- 5: **for** iteration step $n = 1, 2, \dots, N$ **do**
- 6: Reset state s_0 .
- 7: **for** timestep in the environment $t = 1, 2, \dots, T$ **do**
- 8: Select action using Eq. (18):
 $a_t = \pi(s_t; \theta) + \tilde{k}\beta$.
- 9: Sample a transition from the environment:
 $s_{t+1}, r_t, d_t \sim p(s_{t+1}|s_t, a_t)$.
- 10: Store the transition in the replay buffer:
 $\mathcal{D} \leftarrow \mathcal{D} \cup \{(s_t, a_t, r_t, s_{t+1}, d_t)\}$.
- 11: **end for**
- 12: **for** gradient step $g = 1, 2, \dots, G$ **do**
- 13: Sample mini-batch of transitions from replay buffer $\mathcal{D}$.
- 14: Approximate the optimal adversarial observation perturbations through Algorithm 1:

$$\Delta^* \leftarrow \begin{cases} \max & \mathbb{E}[\|\pi(a|s) - \pi(\tilde{a}|s + \Delta)\|_2], \\ \text{s.t.} & |\Delta| \leq \delta. \end{cases}$$
- 15: Update the actor network parameters using Eq. (31):
 $\theta \leftarrow \nabla_{\theta} J_{\pi}(\theta)$.
- 16: Update the critic network parameters using Eq. (17):
 $\phi^1 \leftarrow \nabla_{\phi^1} J_c(\phi^1)$, $\phi^2 \leftarrow \nabla_{\phi^2} J_c(\phi^2)$.
- 17: Update the dual variables using Eq. (32):
 $\lambda_1, \lambda_2 \leftarrow \nabla_{\lambda_1, \lambda_2} J_d(\lambda_1, \lambda_2)$.
- 18: Update the target action-value function network parameters using Eq. (23):
 $\bar{\phi}^1 \leftarrow \rho\bar{\phi}^1 + (1 - \rho)\phi^1$, $\bar{\phi}^2 \leftarrow \rho\bar{\phi}^2 + (1 - \rho)\phi^2$.
- 19: **if** $g \bmod \omega$ **then**
- 20: Update the adversary network parameters using Eq. (30):
 $\bar{\theta} \leftarrow \nabla_{\bar{\theta}} J_{\bar{\pi}}(\bar{\theta})$.
- 21: **end if**
- 22: **end for**
- 23: **end for**

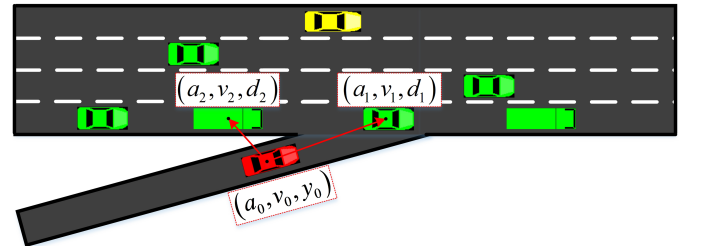


Fig. 2. Illustration of states adopted by autonomous driving RL agent.

To implement the highway on-ramp merging decision making scheme based on the proposed CARL, the state, action and reward of the autonomous driving agent needs to be defined. The states (i.e., inputs) of the agent include 9 dimensions, and the detailed description is given in Table I and Fig. 2.

TABLE I
STATES OF AUTONOMOUS DRIVING RL AGENT

Parameters (Unit)	Definition
a_0 (m/s ²)	Longitudinal acceleration of the autonomous vehicle
v_0 (m/s)	Velocity of the autonomous vehicle
y_0 (m)	Position of the autonomous vehicle on Y-axis
a_1 (m/s ²)	Longitudinal acceleration of the vehicle in front
v_1 (m/s)	Velocity of the vehicle in front
d_1 (m)	Distance from the vehicle in front
a_2 (m/s ²)	Longitudinal acceleration of the vehicle behind
v_2 (m/s)	Velocity of the vehicle behind
d_2 (m)	Distance from the vehicle behind

Moreover, the action (i.e., output) of our agent is a continuous longitudinal acceleration or deceleration. According to the presets of the passenger vehicle in SUMO, the extremum acceleration and deceleration of the autonomous vehicle are 2.60 m/s² and -9.00 m/s², respectively.

Algorithm 3 Reward Function Design for RL Agent

Input: State and action of RL agent.

```

1:  $r(s, a) = v_0/50$ .  $\triangleright$  Encourage the agent to be more efficiency
2: if  $1.47 < a_0$  or  $a_0 < -2.00$  then
3:    $r(s, a) = r(s, a) - 0.10$ .  $\triangleright$  Penalize drastic speed changes
4: end if
5: if Collision occurs then
6:    $r(s, a) = r(s, a) - 0.50$ .  $\triangleright$  Penalize collision
7: else
8:   if Reach the target lane then
9:      $r(s, a) = r(s, a) + 0.20$ .  $\triangleright$  Lead to the target lane
10:  end if
11: end if

```

Output: $r(s, a)$.

Since the reward function plays a crucial role for learning the robust highway on-ramp merging decision making behaviors, it is designed by considering safety, comfort and transport efficiency. In terms of autonomous driving comfort, we employ the longitudinal acceleration setting for normal driver, aggressive driver and extremely aggressive driver in [43]. For instance, when the longitudinal acceleration of our autonomous vehicle exceed 1.47 m/s² or its longitudinal deceleration is less than -2 m/s², the autonomous driving RL agent will be subject to certain penalties. Algorithm 3 describes the design of the reward function in detail.

The adversary, actor and critic networks are designed through two fully connected hidden layer, and the size of the both of hidden layers is 128. Moreover, all activation functions in hidden layers are ReLU. The dimensions of input and output of the neural networks are 9 and 1 respectively. The main hyperparameters of the proposed approach are given in Table II.

V. TESTING RESULTS AND PERFORMANCE EVALUATION

A. Environment

The simulation experiment with SUMO platform is implemented to evaluate the performance of the proposed robust

TABLE II
THE MAIN HYPERPARAMETERS OF THE PROPOSED ALGORITHM

Parameters	Value	Parameters	Value
Discount factor γ	0.99	Scale coefficient ρ	0.995
Adversary learning rate $l_{\bar{a}}$	0.0001	Actor learning rate l_a	0.0001
Dual learning rate l_d	0.0001	Critic learning rate l_c	0.001
Attack learning rate l_{Δ}	0.10	Constraint threshold ϵ_1	0.05
Constraint threshold ϵ_2	0.05	Adversarial weight η	0.10
Upper bound k	1.00	Weight coefficient $\tilde{k}$	0.01

TABLE III
FINAL PERFORMANCE OF DIFFERENT MEHRODS IN MODEL TRAINING

	Average Return	Average Speed	TTSTL
PPO	8.57 $\pm$ 3.93	18.83 $\pm$ 1.50	12.21 $\pm$ 5.38
TD3	14.98 $\pm$ 1.37	22.23 $\pm$ 0.14	22.63 $\pm$ 2.55
SAC	14.24 $\pm$ 0.43	21.75 $\pm$ 0.47	25.07 $\pm$ 0.33
CARL	17.80 $\pm$ 0.58	22.63 $\pm$ 0.51	25.79 $\pm$ 0.28

highway on-ramp merging decision making method for autonomous vehicles. SUMO is leveraged to create three stochastic mixed traffic flows with different densities at highway on-ramp scenarios. The ego and social vehicles can perform lane change maneuvers via the LC2013 model in SUMO [44]. The speed of social vehicles is controlled by the intelligent driving model (IDM) of SUMO.

Fig. 3 illustrates our test scheme. The traffic environment consists of a four-lanes-based highway and a ramp. The target lane of all vehicles in each lane is the lane that intersects with the ramp. In addition, P_n , P_l and P_h are denoted as the probabilities for emitting a vehicle each second in mixed traffic flows with normal, low and high densities respectively. Moreover, P_n , P_l and P_h are set as 0.80, 0.60, 1.00 respectively. For each testing case, five stochastic traffic flows are generated by SUMO. The maximum traffic speed of four of all lanes is set to 30 m/s, and the maximum speed of one of them is 35 m/s.

The proposed approach and baseline approaches is evaluated in both training and testing. The policy models are trained and tested in the mixed traffic flow with normal density. At the same time, the mixed traffic flows with low and high densities are only employed to test the policy models.

As shown in Fig. 4, unlike the model training stage using the RL algorithms, in model testing, the autonomous driving RL agent observes the state $\tilde{s}_t$ perturbed by white-box attack technique rather than the state s_t . Additionally, the policies for interacting with the environment contain adversary attacks.

B. Baseline

The PPO [45] based autonomous driving on-ramp merging decision making method is implemented as a classical baseline. Moreover, since Twin Delayed DDPG (TD3) [41] and soft actor-critic (SAC) [46], [47] are state-of-the-art RL algorithm, they are leveraged as state-of-the-art baselines.

C. Evaluation

Each evaluation calculates the agents' average return, speed and travel time steps in the target lane over 50 episodes. The maximum time step to perform a merging task is set as 30

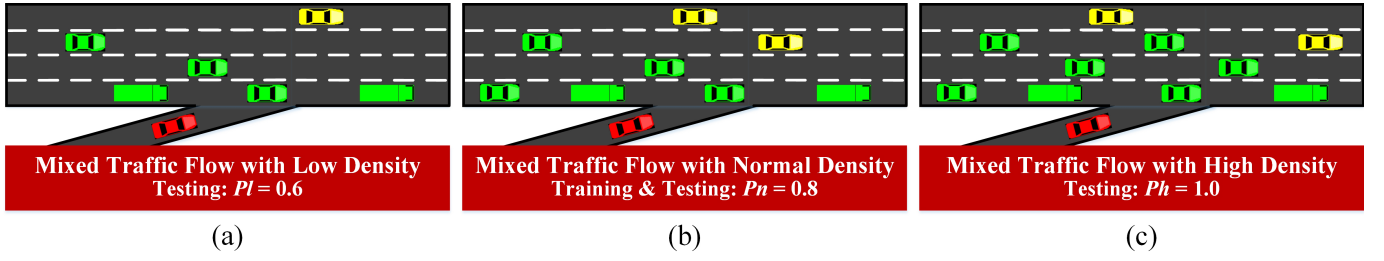


Fig. 3. Experimental environment via SUMO-based mixed traffic flow with a random number of vehicles. (a) Mixed traffic flow with low density; (b) Mixed traffic flow with normal density; (c) Mixed traffic flow with high density. The target lane for all vehicles is the lane that intersects with the ramp.

TABLE IV
EVALUATION OF THE DRIVING POLICIES TRAINED VIA DIFFERENT ALGORITHMS IN THREE STOCHASTIC MIXED TRAFFIC FLOWS

Environment	Metric	PPO	TD3	SAC	CARL
Low Density	Utility	8.74 ± 2.86	11.05 ± 3.46	11.54 ± 1.82	14.67 ± 3.88
	Robustness	6.38 ± 1.90	6.80 ± 2.21	7.44 ± 0.49	3.93 ± 1.15
	Speed (m/s)	13.07 ± 6.81	18.33 ± 7.31	17.70 ± 5.98	19.80 ± 4.64
	Success Rate	0.58 ± 0.18	0.75 ± 0.18	0.70 ± 0.13	0.90 ± 0.11
	Collision Times	1.05 ± 1.42	0.51 ± 1.12	1.04 ± 1.43	0.31 ± 0.91
Normal Density	Utility	8.78 ± 2.99	11.68 ± 3.74	12.47 ± 0.68	14.94 ± 3.98
	Robustness	6.33 ± 1.83	6.33 ± 2.68	7.60 ± 0.01	3.09 ± 1.41
	Speed (m/s)	13.12 ± 6.85	18.75 ± 7.47	17.81 ± 5.81	19.93 ± 4.43
	Success Rate	0.57 ± 0.19	0.77 ± 0.19	0.74 ± 0.02	0.92 ± 0.13
	Collision Times	1.07 ± 1.43	0.36 ± 0.97	0.78 ± 1.31	0.26 ± 0.84
High Density	Utility	9.10 ± 3.25	11.33 ± 3.10	11.80 ± 1.35	14.46 ± 4.24
	Robustness	6.30 ± 1.76	6.65 ± 2.29	7.45 ± 0.46	3.53 ± 2.13
	Speed (m/s)	13.26 ± 6.96	18.51 ± 7.11	17.79 ± 6.00	19.66 ± 4.84
	Success Rate	0.59 ± 0.19	0.77 ± 0.16	0.71 ± 0.12	0.88 ± 0.14
	Collision Times	1.00 ± 1.41	0.46 ± 1.08	0.97 ± 1.40	0.35 ± 0.96

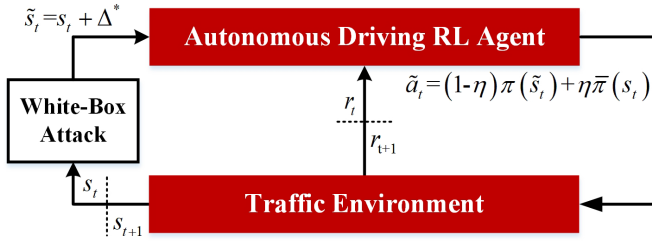


Fig. 4. Schematic diagram of model evaluation method. In model testing, the autonomous driving RL agent observes the perturbed state $\tilde{s}_t$ rather than the state s_t . Meanwhile, the policies for interacting with the environment contain adversary attacks.

(i.e., 1 episode). Fig. 5 shows the results of PPO, TD3, SAC and CARL algorithms during model training in the highway on-ramp merging scenario based on stochastic mixed traffic flow with normal density. The final performance of all the methods is provided in Table III. Bold number is the best in each column of Table III. These four schemes are evaluated for five trials by different random seeds in the stochastic mixed traffic flow with normal density. Moreover, the solid curve corresponds to the mean, and the shaded region represents the standard deviation.

Fig. 5 and Table III demonstrate that the proposed robust highway on-ramp merging decision making approach with CARL outperforms the baseline methods with a large margin, both in terms of the final performance and the learning efficiency. For instance, compared with the PPO, TD3 and SAC methods, CARL gains 107.70%, 18.83% and 25.00% improvements with respect to the final return respectively. It can be seen that, CARL agent is superior to PPO, TD3 and SAC agents in terms of the final driving speed. Additionally,

in contrast to the PPO, TD3 and SAC algorithms, the average travel time steps of CARL driving agent in the target lane is enhanced by about 111.22%, 13.96% and 2.87% respectively. It means that the CARL autonomous driving agent can reach the target lane faster than the vehicles based on other methods while avoiding collision. In other words, under fixed training time steps, our autonomous vehicle can reach the target lane quickly and safely and drive longer in the target lane.

To evaluate the robustness and performance of the learned policy under the adversarial observation perturbations and the adversarial environmental disturbances, we test each RL policy model in the 121 different scenarios with jointly varying both the upper bound k and the adversarial weight η . 30 time steps are used for one test under a fixed k and a fixed η . Then, the cumulative value of the reward function r of a RL agent is calculated under 30 time steps. Furthermore, the mean and the standard deviation of the average cumulative reward for a RL agent in the 121 scenarios is adopted to quantify the utility and the robustness of its policies under environmental uncertainties respectively. This means the policy with smaller standard deviation of the average cumulative reward shows stronger robustness. Additionally, we report the merging success rate and collision times of each autonomous driving agent in the 121 scenarios.

For each agent, we evaluate 5 final policy models trained with different random seeds and report the average metrics in Table IV. The performance of the CARL policy outperforms PPO, TD3 and SAC policies in three stochastic mixed traffic flows based on different densities in terms of all the metrics. For example, in comparison with the PPO, TD3 and SAC policies, the CARL policy gains 67.85%, 32.76% and 27.12%

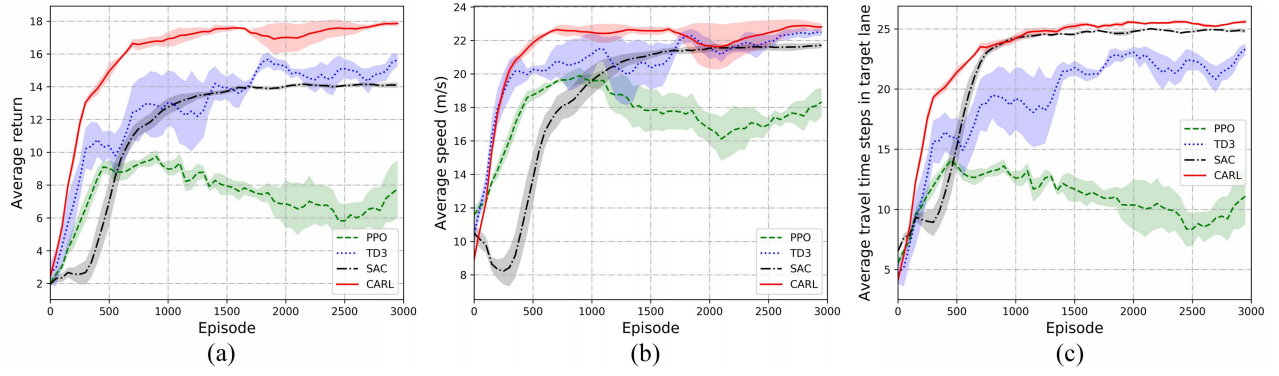


Fig. 5. Training curves by the PPO, TD3, SAC and CARL algorithms in SUMO based mixed stochastic traffic flow with normal density. The maximum length of each episode is 30 time steps. (a): Average return; (b): Average speed; (c): Travel time steps in the target lane (TTSTL).

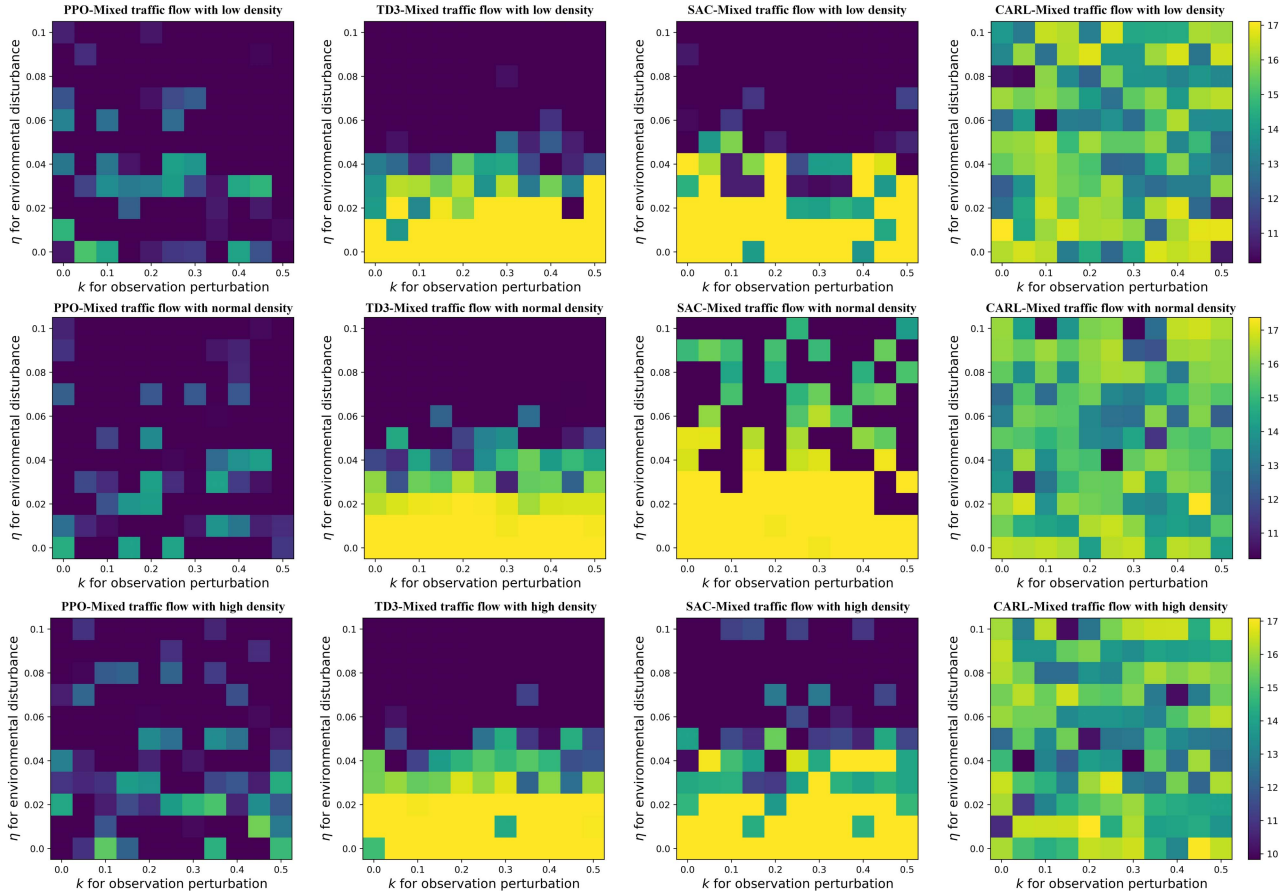


Fig. 6. Evaluation heatmap for PPO, TD3, SAC and CARL agents in stochastic mixed traffic flows with different densities under environmental uncertainties.

improvements with respect to utility in mixed traffic flow with low density respectively. Compared with the PPO, TD3 and SAC policies, the robustness of the CARL policy is improved by about 38.40%, 42.21% and 47.18% respectively. In contrast to the PPO, TD3 and SAC policies, the CARL policy gains 55.17%, 20.00% and 28.57% improvements with respect to merging success rate in mixed traffic flow with low density respectively.

In the stochastic mixed traffic flow scenario based on normal density, the utility of the OARL policy is superior to one of PPO, TD3 and SAC policies. Moreover, compared with the PPO, TD3 and SAC policies, CARL policy gains 51.19%, 51.19% and 59.34% improvements with respect to robustness respectively. In comparison with the PPO, TD3 and SAC

policies, the merging success rate of the CARL policy is enhanced by about 61.40%, 19.48% and 24.32% respectively. In contrast to the PPO, TD3 and SAC policies, the collision times of the CARL policy are reduced by about 75.70%, 27.78% and 66.67% in mixed traffic flow with normal density respectively.

In the stochastic mixed traffic flow scenario based on high density, compared with the PPO, TD3 and SAC agents, the speed of the CARL agent is improved by about 48.27%, 6.21% and 10.70% respectively. In comparison with the PPO, TD3 and SAC agents, the merging success of the CARL agent is enhanced by about 49.15%, 14.29% and 23.94% respectively. Additionally, compared with the PPO, TD3 and SAC agents, the collision times of the CARL agent are reduced by about

65.00%, 23.91% and 63.91% respectively. Therefore, it can be seen that the proposed scheme performs consistently in three different highway on-ramp merging scenarios.

In addition, Fig. 6 visually illustrates the performance of PPO, TD3, SAC and CARL policies in the stochastic mixed traffic flows with low, normal and high densities. A yellow lattice denotes a high value of the average cumulative reward, while a dark lattice denotes the reverse. The colors of the lattices look similar, which makes the heatmap approximate solid color, and the corresponding policy has better robustness. It can be found that in contrast to PPO, TD3 and SAC policies, the heatmap color with CARL policy is more uniform. Furthermore, the performance of CARL policy is scarcely influenced by the adversarial observation perturbations and the adversarial environmental disturbances. This means that the proposed robust highway on-ramp merging decision making method with OARL can improve the the performance of autonomous driving RL agent while guaranteeing the robustness of the policy against observation uncertainties.

VI. CONCLUSION

This paper proposes a novel CARL approach for robust highway on-ramp merging decision making of autonomous vehicles. Environmental disturbance is modelled as an adversarial agent that can learn an optimal adversarial policy to thwart the goal of the autonomous driving agent. Meanwhile, observation perturbation is approximated to maximize the variation of the perturbed policy via a white-box adversarial attack technique. Furthermore, a CAAC algorithm is introduced to optimize an on-ramp merging policy while keeping the variations of the attacked driving policy and action-value function within bounds.

The evaluation results in three stochastic mixed traffic flows based on different densities demonstrate that our autonomous agent can make merging decisions robustly and safely at highway on-ramps under adversarial environmental uncertainties. Additionally, in contrast to the three baseline schemes, the policy models trained by the proposed algorithm show superior performance and robustness. Our autonomous agent can also provide more stable performance than the baselines in terms of speed, success rate and collision times.

Although the proposed method shows great potential, limitations still remain. Firstly, SUMO can provide diverse traffic flows in simulations, however, there is still a gap between IDMs and human drivers in terms of driving behavior. Secondly, the proposed approach considers the worst-case setting, however, how to certify its robustness with theoretical guarantees is still an opening question and needs to be investigated. Consequently, future work involves model fine-tuning by using merging data collected in real world. Moreover, a certified CARL solution will be investigated to further improve the trustworthiness of autonomous driving decision-making.

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